About 2,600 results for “review wildfire spread prediction”

Fields of Study | Date Range | Has PDF | Author | Journals & Conferences | More Filters | Filters | Sort by Relevance

Modeling wildfires' spread based on low-cost sensors measuring real-time moisture content

Nikolaos Avgoustis | Alvertos Ioannis Mourikis | Marios N. Anagnostou | Ioannis Karydis | M. Avlonitis

Environmental Science, Engineering · International Conference on Information and… · 2024

This study addresses the growing challenge of accurate **wildfire prediction** by integrating low-cost, real-time sensors into fire **spread** simulations. Traditional models, which often rely on static… Expand

PDF | View Paper | Save | Cite

Future IoT tools for COVID-19 contact tracing and prediction: A review of the state-of-the-science

V. Jahmunah | Vidya K. Sudarshan | +8 authors | U. R. Acharya

Computer Science, Environmental Science · International journal of imaging systems and… · 9 February 2021

TLDR A literature **review** of contact tracing methods and applications is conducted, found limitations in gathering health data, such as insufficient network coverage, and proposes a novel intelligent tool that will be useful for contact tracing and **prediction** of COVID-19 clusters. Expand

52 | PDF | Wiley | Save | Cite

Machine Learning based Wildfire Area Estimation Leveraging Weather Forecast Data

Saket Sultania | Rohit Sonawane | Prashasti Kanikar | Environmental Science, Computer Science · International Journal of Information Technology… · 8 February 2025

TLDR This research presents a framework that integrates satellite remote sensing and numerical weather **prediction** model data to refine estimations of final **wildfire** sizes, and optimize an ensemble model by comparing various algorithms to minimize the root mean squared error between the predicted and actual burned areas. Expand

1 | Publisher | Save | Cite

Wildfire Detection Based on Satellite Images Using Deep Learning

Zehui Wen | Feng Gu | Environmental Science, Computer Science · 2025 8th International Conference on Artificial… · 23 May 2025

TLDR This paper proposes a deep learning method using Convolutional Neural Networks (CNNs) to predict the **wildfire** based on the satellite image data set, and results show the achieved **prediction** accuracy reaches up to 96%, validating the effectiveness of the proposed model. Expand

1 | IEEE | Save | Cite

Using Satellite-Derived Fire Arrival Times for Coupled Wildfire-Air Quality Simulations at Regional Scales of the 2020 California Wildfire Season

W. Lassman | J. Mirocha | +6 authors | M. Barbato | Environmental Science · Journal of Geophysical Research: Atmospheres · 10 January 2023

Wildfire frequency has increased in the Western US over recent decades, driven by climate change and a legacy of forest management practices. Consequently, human structures, health, and life are… Expand

6 | PDF | Wiley | Save | Cite

Deep Learning and Transformer Approaches for UAV-Based Wildfire Detection and Segmentation

Rafik Ghali · M. Akhloufi · Wided Souidène Mseddi · Environmental Science, Computer Science · Italian National Conference on Sensors · 1 March 2022

TLDR A novel deep ensemble learning method, which combines EfficientNet-B5 and DenseNet-201 models, is proposed to identify and classify **wildfire** using aerial images and proved its ability in classifying **wildfire** even small fire areas. Expand

112 [PDF] PDF Save Cite

Indices for the evaluation of wildfire spread simulations using contemporaneous predictions and observations of burnt area

T. Duff · D. M. Chong · K. Tolhurst · Environmental Science · Environmental Modelling & Software · 1 September 2016

21 Publisher Save Cite

Generative Algorithms for Wildfire Progression Reconstruction from Multi-Modal Satellite Active Fire Measurements and Terrain Height

B. Shaddy · Brianna Binder · +8 authors · Assad A. Oberai

Environmental Science, Computer Science · arXiv.org · 12 June 2025

TLDR An approach is developed for estimating fire progression from VIIRS active fire measurements, GOES-derived ignition times, and terrain height data, and it is observed that terrain has minimal influence when the inference is conditioned on satellite measurements. Expand

[PDF] arXiv Save Cite

A Semi-Supervised Wildfire Image Segmentation Network with Multi-Scale Structural Fusion and Pixel-Level Contrastive Consistency

Yong Sun · Wei Wei · Jia-Nan Guo · Haifeng Lin · Yiqing Xu

Environmental Science, Computer Science · Fire · 7 August 2025

TLDR This work proposes a semi-supervised **wildfire** image segmentation framework that enhances segmentation performance under limited annotation conditions by integrating multi-scale structural information fusion and pixel-level contrastive consistency learning. Expand

Publisher Save Cite

Dynamic UAV Path Planning Under Wildfire Scenarios Using PPO Algorithm

Xinlong Cai · Wenjie Tang · Environmental Science, Engineering · 2025 4th International Conference on Robotics… · 4 July 2025

TLDR This work proposes an algorithm that integrates fire **spread prediction** with reinforcement learning (RL)-based path planning that enables adaptive trajectory optimization that dynamically responds to **wildfire** evolution. Expand

IEEE Save Cite

← … 15 16 17 18 … →

Stay Connected With Semantic Scholar

Your E-mail Address Sign Up

Semantic Scholar is a free, AI-powered research tool for scientific literature, based at Ai2.

Learn More

About Us
Publishers
Blog
Ai2 Careers

Product Overview
Semantic Reader
Scholar's Hub
Beta Program
Release Notes

API Overview
API Tutorials
API Documentation
API Gallery

Publications
Research Careers
Resources

FAQ
Librarians
Tutorials
Contact

Proudly built by Ai2
Collaborators & Attributions • Terms of Service • Privacy Policy • API License Agreement

Ai2